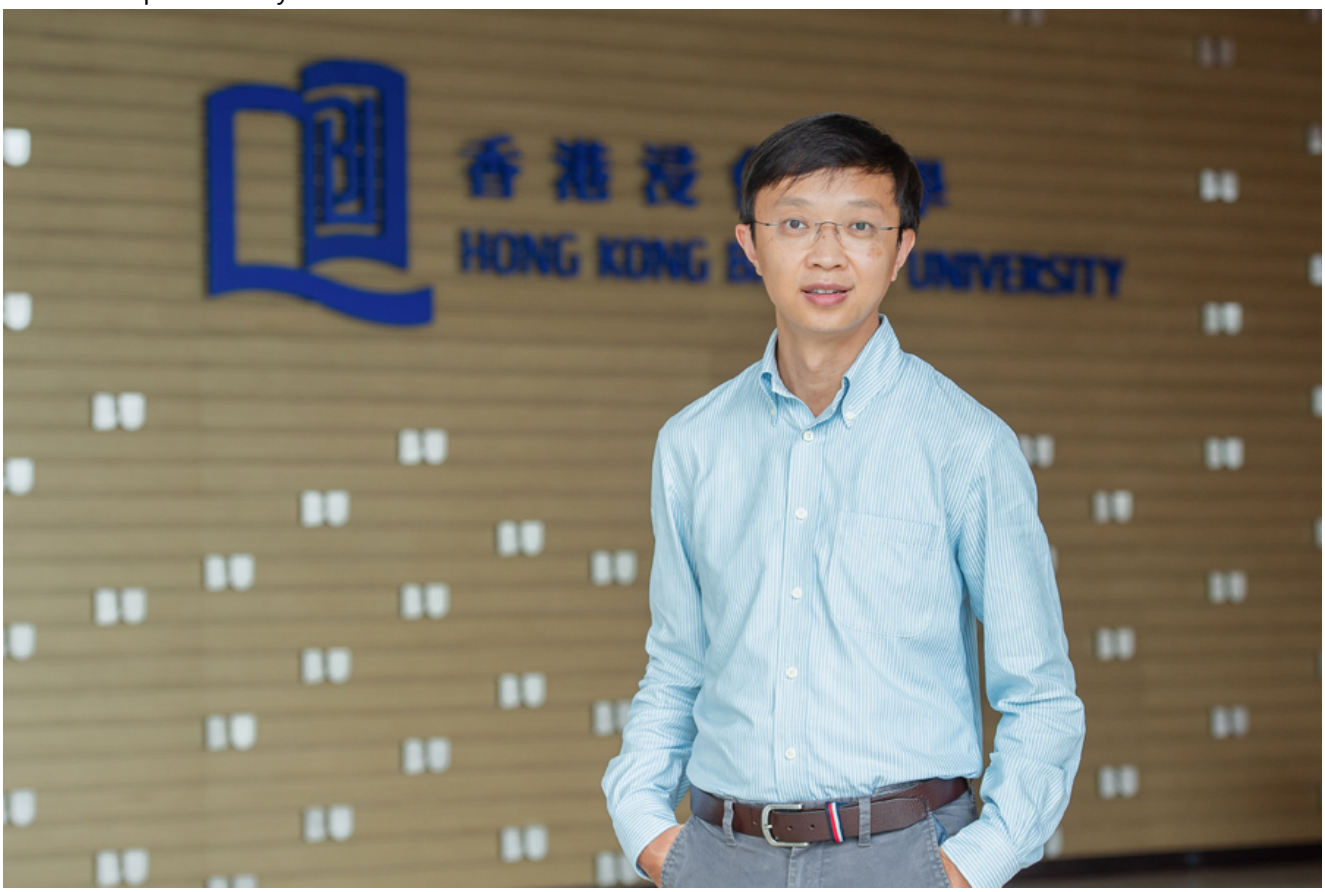

Prof. DAI, Henry Hong Ning

戴弘寧教授

BEng, MEng, DEng, PhD
Associate Professor, Department of Computer Science

henrydai@comp.hkbu.edu.hk

DLB 643 +852 3411 8015

Personal Webpage HKBU Scholars

About

Dr. Hong-Ning Dai (Henry) obtained a Ph.D. in Computer Science and Engineering from the Department of Computer Science and Engineering at the Chinese University of Hong Kong and a D.Eng. in Computer Technology Application from the Department of Computer Science and Engineering at Shanghai Jiao Tong University. Before joining HKBU, he worked in the Department of Information Engineering at the Chinese University of Hong Kong, the School of Computer Science and Engineering at Macau University of Science and Technology, and the Department of Computing and Decision Sciences at Lingnan University, Hong Kong. He has published more than 300 papers in refereed journals and conferences, including Proceedings of the IEEE, IEEE JSAC, IEEE TMC, IEEE TPDS, IEEE Trans. on Computer, IEEE TSE, IEEE TDSC, IEEE TKDE, IEEE TNNLS, IEEE/ACM, ICSE, ACM/IEEE ASE, IEEE INFOCOM, ICDE, ICDCS, AAAI, ACM CSUR, ACM TOSEM, IEEE Communications Surveys & Tutorials, etc. He is a senior member of ACM, IEEE, and EAI. He has served as an associate editor for IEEE Communications Survey and Tutorials, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Industrial Informatics, IEEE Transactions on Industrial Cyber-Physical Systems, Ad Hoc Networks, and Connection Science. He has been a PC member of many top-tier conferences. He has also served as general chair/PC chair of many international conferences.

Dr. Henry Dai is recruiting **self-motivated Ph.D./RA/Post-doc** with a strong background in computer science, electronic engineering, or applied mathematics, working in fields including (but not limited to) security and reliability of VR/AR systems, blockchain, and the Internet of Things. More details about Ph.D. applications and Ph.D. scholarships are as follows.

- Ph.D. Application
- Hong Kong Ph.D. Fellowship Scheme (HKPFS)

Research Interests

- Security and Reliability of VR/AR Systems
- Internet of Things and Mobile Sensing
- Blockchain and Distributed Systems

Selected Publications

- Jingwen Wu, Hanyang Guo, **Hong-Ning Dai**, and Xiapu Luo, “XRFix: Exploring Performance Bug Repair of Extended Reality Applications with Large Language Models”, in the 48th IEEE/ACM International Conference on Software Engineering (**ICSE**), Rio de Janeiro, Brazil, 2026
- Zhengyang Zhu, **Hong-Ning Dai**, Hanyang Guo, Zeqing Liao, and Zibin Zheng, “VRExplorer: A Model-based Approach for Semi-Automated Virtual Reality Scene Testing”, in the 40th IEEE/ACM International

Conference on Automated Software Engineering (**ASE**), Seoul, South Korea, 2025

- Fengliang He, **Hong-Ning Dai**, Hanyang Guo, Xiapu Luo, and Jiadi Yu, “AcouListener: An Inaudible Acoustic Side-channel Attack on AR/VR Systems”, in the 30th European Symposium on Research in Computer Security (**ESORICS**), Toulouse, France, 2025
- Jian Zheng, Huawei Huang, Yinqiu Liu, Taotao Li, **Hong-Ning Dai**, and Zibin Zheng, “Justitia: An Incentive Mechanism towards the Fairness of Cross-shard Transactions”, in the IEEE Conference on Computer Communications (**INFOCOM**), London, United Kingdom, 2025
- Yunzhong Chen, Jiadi Yu, Hao Kong, Daqiang Zhang, Yanmin Zhu, **Hong-Ning Dai**, Yi-Chao Chen, and Linghe Kong, “EMTap: Eavesdropping on Sound Transmitted via Audio Cables by Sensing Electromagnetic Variation using RFID”, in ACM international joint conference on Pervasive and Ubiquitous Computing (**UbiComp**), Espoo, Finland, 2025
- Junlin Yang, Jiadi Yu, Linghe Kong, Yanmin Zhu, and **Hong-Ning Dai**, “Toward Open-World-Aware User Authentication Based on Human Bodies Using mmWave Signals”, in IEEE Transactions on Mobile Computing (**TMC**), vol. 24, no. 9, pp. 9036-9049, Sept. 2025
- Hanyang Guo, **Hong-Ning Dai**, Xiapu Luo, Zibin Zheng, Genyang Xu, and Fengliang He, “An Empirical Study on Oculus Virtual Reality Applications: Security and Privacy Perspectives”, in the 46th International Conference on Software Engineering (**ICSE**), Lisbon, Portugal, 2024.
- Wuhui Chen, Yikai Feng, Jianting Zhang, Zhongteng Cai, **Hong-Ning Dai**, and Zibin Zheng, “Auncel: Fair Byzantine Consensus Protocol with High Performance”, in IEEE Conference on Computer Communications (**INFOCOM**), Vancouver, Canada, 2024.
- Zhijie Zhong, Zibin Zheng, **Hong-Ning Dai**, Qing Xue, Junjia Chen, and Yuhong Nan, “PrettySmart: Detecting Permission Re-delegation Vulnerability for Token Behaviors in Smart Contracts”, in the 46th International Conference on Software Engineering (**ICSE**), Lisbon, Portugal, 2024.
- Junqin Huang, Linghe Kong, Long Cheng, **Hong-Ning Dai**, Meikang Qiu, Guihai Chen, Xue Liu, and Gang Huang, “Block-Sense: Towards Trustworthy Mobile Crowdsensing via Proof-of-Data Blockchain”, in IEEE Transactions on Mobile Computing (**TMC**), vol. 23, no. 2, pp. 1016-1033, 2024.